

Revision Notes for Class 10 Science

Chapter 1 – Chemical Reactions and Equations

Chemical Change: A change that results in the formation of one or more new compounds. Chemical changes are also known as chemical reactions.

Observations in a Chemical Reaction

In a chemical reaction or chemical change, the following observations can be made,

- Formation of new substances.
- Change in mass.
- Changes in energy.
- Evolution of gas.
- Change in temperature.
- Permanent change in colour and/or state.
- Formation of a precipitate.

Examples :

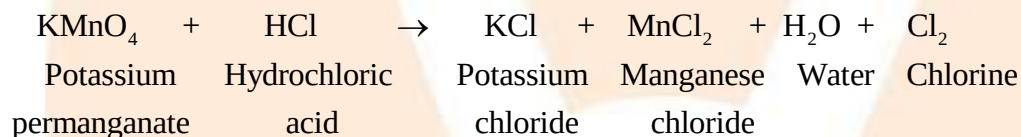
- A. Cooking of food
- B. Rusting of iron
- C. Heating of Lead nitrate
- D. Souring of milk
- E. Ripening of fruit.

Chemical Formulae - A substance's chemical formula is a symbolic representation of the number of atoms present in a single molecule of that material.

- One water molecule has the formula H_2O . It demonstrates that one molecule of water is made up of two hydrogen atoms and one oxygen atom.
- One molecule of sulphuric acid has the formula H_2SO_4 . It illustrates that one molecule of sulphuric acid has 2 hydrogen atoms, 1 sulphur atom, and 4 oxygen atoms.

Chemical Equations - The chemical equation of the reaction is the representation of a chemical change in terms of symbols and formulae of the reactants and products.

Example:



Steps for writing Chemical Equation

A. Step I: Writing Skeletal Equation

A word-equation depicts the transformation of reactants into products by placing an arrow between them ($\rightarrow$). The arrow indicates the reaction's direction and might be read as "to yield," "to form," or "to give."

The reactants are on the left (LHS) of the arrow, while the products are on the right (RHS). A plus sign (+) links the different reactants and products together.

A balanced account of a chemical transaction is a complete chemical equation, which symbolically depicts the reactants, products, and their physical states.

B. Step II: Balancing of Chemical Equation

A balanced chemical equation is one in which the number of atoms of each element is equal on both sides of the equation.

On the reactant and product sides of the equation, a chemical equation is balanced if the number of atoms of each type involved in the reaction is the same.

Equations must be balanced at all times.

The amount of hydrogens on both sides should be an even number whenever H_2O is present on one side (2 atoms of hydrogen in water). If there are 4 H_2O hydrogen atoms, then $4 \times 2 = 8$ hydrogen atoms are present. If there are 7 H_2O hydrogen atoms, then $7 \times 2 = 14$ hydrogen atoms, indicating that all the numbers are even.

There must be an even number in front of HCl on the reactant side. (We'll figure out what that number is later.) As a result, there will be an even amount of chlorine atoms (i.e., $\text{KCl}=1$, $\text{MnCl}_2 = 2$, $\text{Cl}_2 = 2$. i.e., $1 + 2 + 2 = 5$). However, the quantity of chlorine atoms in the product is unusual. KCl has the only odd number of chlorine atoms. Let's alter it to the simplest even number we can think of, which is 2.



Since the number of K atoms in $2 \text{KCl} = 2$, we have to put 2 in front of KMnO_4 for balancing K atom.



In 2KMnO_4 , there are 2 K, 2 Mn, and 8 O. So, add these numbers in front of K, Mn and O, (K is already done).



If there are 8 H_2O on the product side, there should be 16 H (8×2) on the reactants side as well.



Now the only unbalanced one is Chlorine. On the left hand side, there are 16 Cl. On the right hand side, firstly, there are 2 Cl in 2 KCl + 4 Cl in 2 MnCl₂, making total of 6(2 + 4). So, 10 more Cl atoms are to be accounted for. So, place 5 in front of Cl₂ to make it 10 (5 x 2).



The Hit and Trial approach is a technique for balancing chemical equations.

Short Technique for Balancing a Chemical Equation

Let us take the same chemical reaction and try to balance it using a short technique which will save you time and effort in the exam. Let the number of molecules on each side be,



Now comparing LHS and RHS,

$$\text{K: } a = c$$

$$\text{Mn: } a = d$$

$$\text{O: } 4a = e$$

$$\text{H: } b = 2e$$

$$\text{Cl: } b = c + 2d + 2f$$

Let us take $a = 2$, then

$$c = d = a = 2$$

$$b = 2e = 8a = 16$$

$$\text{So, } e = 8$$

$$\text{So, } 16 = 2 + 2(2) + 2(f)$$

$$f = 5$$

Thus, the required equation is: $2\text{KMnO}_4 + 16\text{HCl} \rightarrow 2\text{KCl} + 2\text{MnCl}_2 + 8\text{H}_2\text{O} + 5\text{Cl}_2$.

From a chemical equation we can conclude:

- A. Various reactants and products names
- B. Reactant and product formulae
- C. The number of moles of the reactants and products in relation to each other
- D. Masses of reactants and products relative to each other
- E. Volumes of gaseous reactants and products in relation to each other

In short – the Steps Involved in Balancing a Chemical Equation can be Listed As:

Step 1: Write the skeleton equation correctly.

Step 2: Start with the compound that has the maximum atoms or maximum types of atoms and the atoms present in it are balanced first.

Step 3: Start by balancing elements that only appear once on each side of the arrow. Then, on each side, balance elements that appear more than once.

Step 4: The Elementary substances are balanced last.

Step 5: To make all the coefficients whole numbers, the entire equation is multiplied by a suitable number if necessary.